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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Examinations September-2023
Common to CSE/ISE

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

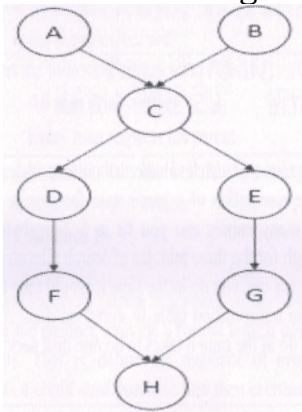
Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

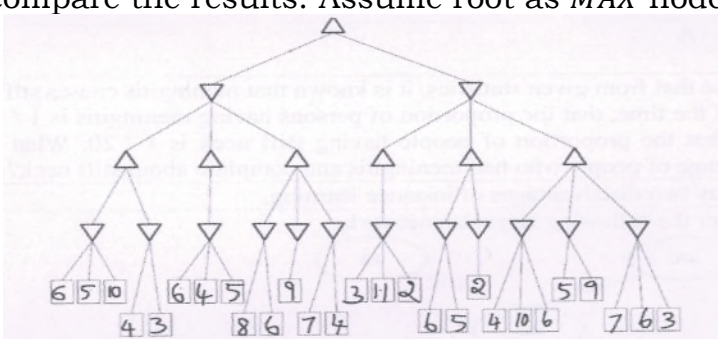
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Suppose that from given statistics, it is known that meningitis causes stiff neck 50% of the time, that the propagation of persons having meningitis is 1/50000, and that the propagation of people having stiff neck is 1/20. What is the percentage of people who had meningitis and complain about stiff neck?	02
	1.2	Give any two disadvantages of instance learning.	02
	1.3	Consider the Bayesian network shown in Fig 1.3:	
		 <pre> graph TD A((A)) --> C((C)) B((B)) --> C((C)) C((C)) --> D((D)) C((C)) --> E((E)) D((D)) --> F((F)) E((E)) --> G((G)) F((F)) --> H((H)) G((G)) --> H((H)) </pre> Fig 1.3	
		a) Are <i>D</i> and <i>E</i> necessarily independent given evidence about both <i>A</i> and <i>B</i> ?	02
		b) Are <i>A</i> and <i>H</i> necessarily independent given evidence about <i>C</i> ?	01
	1.4	State the modus ponens inferencing rule in predicate calculus.	01
	1.5	Define Informed Search Algorithms.	01
	1.6	Which algorithm is used in the Game tree to make decisions of Win/Lose?	01
	1.7	Which AI technique enables the computers to understand the associations and relationships between objects and events?	01
	1.8	Write time and space complexity analysis of <i>BFS</i> ?	02
	1.9	Which term AI describes the common-sense of the judgmental part of problem-solving?	01
	1.10	The search algorithm which is similar to the minimax search, but removes the branches that don't affect the final output is known as _____.	01

1.11	Which process makes two different Logical expressions look identical?	01
1.12	A feature $F1$ can take certain value: A, B, C, D, E & F and represents grade of students from a college. Identify the type of feature type.	01
1.13	List the 4 types informed search.	02
1.14	You are given reviews of few Netflix series marked as positive, negative and neutral. Classifying reviews of a new Netflix series is an example of _____.	01
1.15	_____ is Impact of high variance on the training set	01

PART B

2	a b	In <i>AI</i> define the term agent. Classify different types of agents. Give the working principle of utility-based agent. Consider a scenario you are employing an agent for searching a document from collection of documents, which type of agent you propose?	10 06
3	a b	Apply Minimax algorithm and Alpha-Beta pruning on the game tree shown in Fig 3a and compare the results. Assume root as <i>MAX</i> node.  Fig 3a Suppose you have a chess program that can evaluate 10 million nodes per second. Decide on a compact representation of a game state for storage in a transposition table. About how many entries can you fit in a 2-gigabyte in-memory table? Will that be enough for the three minutes of search allocated for one move? How many table lookups can you do in the time it would take to do one evaluation? Now suppose the transposition table is stored on disk. About how many evaluations could you do in the time it takes to do one disk seek with standard disk hardware? OR A logical knowledge base represents the world using a set of sentences with no explicit structure. An analogical representation, on the other hand, is one in which the representation has structure that corresponds directly to the structure of the thing represented. Consider a road map of your country as an analogical representation of facts about the country. The two-dimensional structure of the map corresponds to the two-dimensional surface of the area. - Give five examples of symbols in the map language. - An explicit sentence is one that the creator of the representation actually writes down. An implicit sentence is one that results from explicit sentences because of properties of the analogical representation. Give three examples each of implicit and explicit sentences in the map language.	10 06
4	a		

	iii) Give three examples of facts about the physical structure of your country that cannot be represented in the map language. iv) Give two examples of facts that are much easier to express in the map language than in first-order logic. v) Give two other examples of useful analogical representations. What are the advantages and disadvantages of each of these languages? b What are the three main components in the constraint satisfaction problem? Explain <i>CSP</i> with Map coloring example.	10 06
5	a Consider the Bayesian network shown in Fig 5a(i), where F = having the flu and C = coughing: <pre> graph LR F((F)) --> C((C)) P_F["P(F) = 0.1"] P_C_F["P(C F) = 0.8"] P_C_not_F["P(C ¬F) = 0.3"] style P_F fill:none,stroke:none style P_C_F fill:none,stroke:none style P_C_not_F fill:none,stroke:none </pre> Fig 5a(i) i) Write down the joint probability table specified by the Bayesian network. ii) Consider the network given in Fig 5a(ii) with the probabilities same as the Fig 5a(i) <pre> graph LR C((C)) --> F((F)) </pre> Fig 5a(ii) Calculate the following: x) $P(C)$ y) $P(F C)$ z) $P(F \neg C)$ iii) Which Bayesian network from the above two would you have specified using the rules learned? b Convert the following sentences to <i>FOL</i>: i) All man drink coffee ii) Every man respects his parent iii) Not all students like both Mathematics and Science iv) Some boys play cricket v) All birds fly OR	08 08
6	a Our description of internet shopping omitted the all-important step of actually buying the product. Provide a formal logical description of buying, using event calculus. That is, define the sequence of events that occurs when a buyer submits a credit-card purchase and then eventually gets billed and receives the product. b We have a bag of three biased coins a, b, and c with probabilities of coming up heads of 20%, 60% and 80%, respectively. One coin is drawn randomly from the bag (with equal likelihood of drawing each of the three coins), and then the coin is flipped three times to generate the outcomes X_1, X_2 and X_3. i) Draw the Bayesian network corresponding to this setup and define the necessary <i>CPTs</i>. ii) Calculate which coin was most likely to have been drawn from the bag if the observed flips come out heads twice and tails once.	06 10

7	a	What is Machine learning? Explain <i>ML</i> with respect to robot driving learning problem in detail.	08
	b	Give decision trees to represent the following boolean functions: i) $A \wedge \neg B$ ii) $A \vee [B \wedge C]$ iii) $A \text{ XOR } B$ iv) $[A \wedge B] \vee [C \wedge D]$	
8	a	Why is <i>KNN</i> a Non Parametric Algorithm? How does the <i>KNN</i> Algorithm Work? Explain with suitable example.	08
	b	What is <i>Q</i> -learning? How can we extend the <i>Q</i> -learning algorithm for the deterministic case to handle nondeterministic <i>MDPs</i> ?	08